



Database Management System Lab (BTCS203P)

B. TECH AI&ML, CYBER SECURITY, Data Science 3RD Semester (Session 2026-2025)

Objective:

To understand and practice SQL Functions.

Prerequisites:

1. Basic knowledge of SQL.
2. MySQL installed and set up.

Lab Sheet 2: SQL Functions

DROP DATABASE IF EXISTS DBMS_Lab3;

CREATE DATABASE DBMS_Lab3;

USE DBMS_Lab3;

```
CREATE TABLE Students (  
    student_id INT PRIMARY KEY,  
    first_name VARCHAR(50),  
    last_name VARCHAR(50),  
    age INT,  
    branch VARCHAR(50),  
    dob DATE  
);  
INSERT INTO Students  
(student_id, first_name, last_name, age, branch, dob)  
VALUES  
(1, 'Rahul', 'Sharma', 20, 'Data Science', '2005-04-15'),  
(2, 'Priya', 'Verma', 21, 'AI & ML', '2004-07-22'),  
(3, 'Aman', 'Gupta', 20, 'Cyber Security', '2005-01-10'),  
(4, 'Neha', 'Singh', 19, 'Data Science', '2006-09-05'),  
(5, 'Riya', 'Mehta', 21, 'AI & ML', '2004-12-18');
```

```
CREATE TABLE Employees (  
    employee_id INT PRIMARY KEY,  
    first_name VARCHAR(50),  
    last_name VARCHAR(50),  
    salary DECIMAL(10,2),  
    joining_date DATE  
);  
INSERT INTO Employees  
(employee_id, first_name, last_name, salary, joining_date)  
VALUES  
(101, 'Amit', 'Kumar', 45678.50, '2022-03-15'),  
(102, 'Sneha', 'Sharma', 52345.75, '2021-07-20'),  
(103, 'Rohit', 'Verma', 38999.40, '2023-01-10'),  
(104, 'Pooja', 'Gupta', 61500.25, '2020-11-05'),  
(105, 'Karan', 'Singh', 47500.80, '2022-09-18');
```

Result Grid						
Filter Rows:						
	student_id	first_name	last_name	age	branch	dob
▶	1	Rahul	Sharma	20	Data Science	2005-04-15
	2	Priya	Verma	21	AI & ML	2004-07-22
	3	Aman	Gupta	20	Cyber Security	2005-01-10
	4	Neha	Singh	19	Data Science	2006-09-05
	5	Riya	Mehta	21	AI & ML	2004-12-18
✱	NULL	NULL	NULL	NULL	NULL	NULL

Result Grid					
Filter Rows:					
	employee_id	first_name	last_name	salary	joining_date
▶	101	Amit	Kumar	45678.50	2022-03-15
	102	Sneha	Sharma	52345.75	2021-07-20
	103	Rohit	Verma	38999.40	2023-01-10
	104	Pooja	Gupta	61500.25	2020-11-05
	105	Karan	Singh	47500.80	2022-09-18
✱	NULL	NULL	NULL	NULL	NULL

A. Number Functions

1. Demonstrate ABS() with suitable examples.

SELECT ABS(-25) AS Absolute_Value;

Result Grid	
Filter Rows:	
	Absolute_Value
▶	25

2. Demonstrate ROUND() with suitable examples.

SELECT ROUND(123.4567, 2) AS Rounded_Value;

Result Grid	
Filter Rows:	
	Rounded_Value
▶	123.46

3. Demonstrate CEIL() and FLOOR().

SELECT

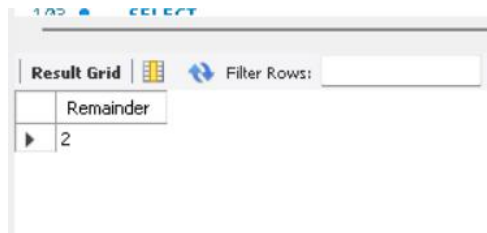
CEIL(12.34) AS Ceiling_Value,

FLOOR(12.89) AS Floor_Value;

Result Grid		
Filter Rows:		
	Ceiling_Value	Floor_Value
▶	13	12

4. Demonstrate MOD() with suitable examples.

```
SELECT MOD(17, 5) AS Remainder;
```

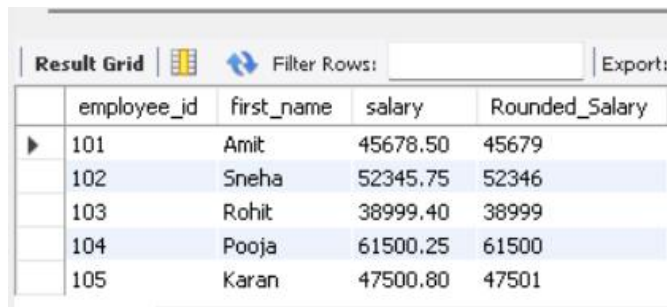


The screenshot shows a database query result in a 'Result Grid' window. The query is 'SELECT MOD(17, 5) AS Remainder;'. The result is a single row with the value '2' in the 'Remainder' column.

Remainder
2

5. Calculate the rounded salary of employees.

```
SELECT  
    employee_id,  
    first_name,  
    salary,  
    ROUND(salary) AS Rounded_Salary  
FROM Employees;
```



The screenshot shows a database query result in a 'Result Grid' window. The query is 'SELECT employee_id, first_name, salary, ROUND(salary) AS Rounded_Salary FROM Employees;'. The result is a table with 5 rows and 4 columns: employee_id, first_name, salary, and Rounded_Salary.

employee_id	first_name	salary	Rounded_Salary
101	Amit	45678.50	45679
102	Sneha	52345.75	52346
103	Rohit	38999.40	38999
104	Pooja	61500.25	61500
105	Karan	47500.80	47501

B. Aggregate Functions

6. Find the total number of students using COUNT().

```
SELECT COUNT(*) AS Total_Students  
FROM Students;
```



The screenshot shows a database query result in a 'Result Grid' window. The query is 'SELECT COUNT(*) AS Total_Students FROM Students;'. The result is a single row with the value '5' in the 'Total_Students' column.

Total_Students
5

7. Find the total salary of all employees using SUM().

```
SELECT SUM(salary) AS Total_Salary  
FROM Employees;
```



The screenshot shows a database query result in a 'Result Grid' window. The query is 'SELECT SUM(salary) AS Total_Salary FROM Employees;'. The result is a single row with the value '246024.70' in the 'Total_Salary' column.

Total_Salary
246024.70

8. Find the average salary using AVG().

```
SELECT ROUND(AVG(salary), 2) AS Average_Salary
FROM Employees;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	Average_Salary			
▶	49204.94			

9. Find the highest salary using MAX().

```
SELECT MAX(salary) AS Highest_Salary
FROM Employees;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	Highest_Salary			
▶	61500.25			

10. Find the lowest salary using MIN().

```
SELECT MIN(salary) AS Lowest_Salary
FROM Employees;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	Lowest_Salary			
▶	38999.40			

C. Character Functions

11. Demonstrate UPPER() on student names.

```
SELECT
    first_name,
    UPPER(first_name) AS Uppercase_Name
FROM Students;
```

Result Grid		Filter Rows:
first_name	Uppercase_Name	
▶ Rahul	RAHUL	
Priya	PRIYA	
Aman	AMAN	
Neha	NEHA	
Riya	RIYA	

12. Demonstrate LOWER() on student names.

```
SELECT
    first_name,
    LOWER(first_name) AS Lowercase_Name
FROM Students;
```

Result Grid			Filter Rows:
	first_name	Lowercase_Name	
▶	Rahul	rahul	
	Priya	priya	
	Aman	aman	
	Neha	neha	
	Riya	riya	

13. Demonstrate LENGTH() on student names.

```
SELECT
    first_name,
    LENGTH(first_name) AS Name_Length
FROM Students;
```

Result Grid			Filter Rows:
	first_name	Name_Length	
▶	Rahul	5	
	Priya	5	
	Aman	4	
	Neha	4	
	Riya	4	

14. Demonstrate CONCAT() using first and last names.

```
SELECT
    first_name,
    last_name,
    CONCAT(first_name, ' ', last_name) AS Full_Name
FROM Students;
```

Result Grid				Filter Rows:
	first_name	last_name	Full_Name	
▶	Rahul	Sharma	Rahul Sharma	
	Priya	Verma	Priya Verma	
	Aman	Gupta	Aman Gupta	
	Neha	Singh	Neha Singh	
	Riya	Mehta	Riya Mehta	

15. Demonstrate SUBSTRING()/SUBSTR() with suitable examples.

```
SELECT
    first_name,
    SUBSTRING(first_name, 1, 3) AS First_Three_Characters
FROM Students;
```

Result Grid			Filter Rows:	Export:
	first_name	First_Three_Characters		
▶	Rahul	Rah		
	Priya	Pri		
	Aman	Ama		
	Neha	Neh		
	Riya	Riy		

D. Conversion Functions

16. Convert a number into a character/string value.

SELECT

```
CAST(12345 AS CHAR) AS Number_As_String;
```



The screenshot shows a database interface with a 'Result Grid' tab. The grid has one column labeled 'Number_As_String' and one row containing the value '12345'. Above the grid, there are icons for 'Filter Rows' and 'Export'. Below the grid, there is a 'Wrap Cell Content' icon.

Number_As_String
12345

17. convert a string into a numeric value.

SELECT

```
CAST('12345' AS UNSIGNED) AS String As Number;
```



The screenshot shows a database interface with a 'Result Grid' tab. The grid has one column labeled 'String_As_Number' and one row containing the value '12345'. Above the grid, there are icons for 'Filter Rows', 'Export', and 'Wrap Cell Content'.

String_As_Number
12345

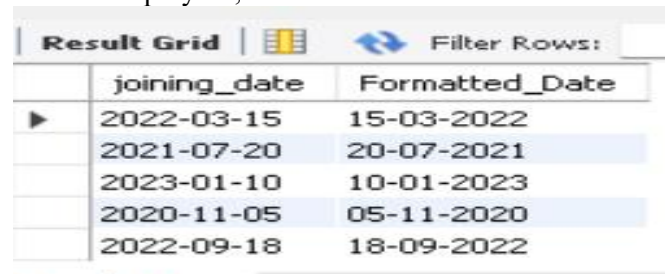
18. Convert a date value into a different format.

SELECT

```
joining_date,
```

```
DATE_FORMAT(joining_date, '%d-%m-%Y') AS Formatted_Date
```

FROM Employees;



The screenshot shows a database interface with a 'Result Grid' tab. The grid has two columns: 'joining_date' and 'Formatted_Date'. It contains five rows of data. Above the grid, there are icons for 'Filter Rows' and 'Export'. Below the grid, there is a 'Wrap Cell Content' icon.

joining_date	Formatted_Date
2022-03-15	15-03-2022
2021-07-20	20-07-2021
2023-01-10	10-01-2023
2020-11-05	05-11-2020
2022-09-18	18-09-2022

19. Demonstrate CAST() with suitable examples.

SELECT

```
CAST(125.75 AS DECIMAL(10,2)) AS Casted_Value;
```



The screenshot shows a database interface with a 'Result Grid' tab. The grid has one column labeled 'Casted_Value' and one row containing the value '125.75'. Above the grid, there are icons for 'Filter Rows', 'Export', and 'Wrap Cell Content'.

Casted_Value
125.75

20. Demonstrate CONVERT() where supported by the DBMS.

SELECT

```
CONVERT('2026-10-04', DATE) AS Converted_Date;
```

E. Date Functions

21. Display the current date using a suitable date function.

```
SELECT  
    CURDATE() AS Current_Date;
```

21. Extract the year, month and day from a date.

```
SELECT  
    joining_date,  
    YEAR(joining_date) AS Joining_Year,  
    MONTH(joining_date) AS Joining_Month,  
    DAY(joining_date) AS Joining_Day  
FROM Employees;
```

Result Grid  Filter Rows: Export: Export				
	joining_date	Joining_Year	Joining_Month	Joining_Day
▶	2022-03-15	2022	3	15
	2021-07-20	2021	7	20
	2023-01-10	2023	1	10
	2020-11-05	2020	11	5
	2022-09-18	2022	9	18

23. Find the difference between two dates.

```
SELECT  
    first_name,  
    joining_date,  
    DATEDIFF(CURDATE(), joining_date) AS Days_Difference  
FROM Employees;
```

Result Grid  Filter Rows:			
	first_name	joining_date	Days_Difference
▶	Amit	2022-03-15	1664
	Sneha	2021-07-20	1902
	Rohit	2023-01-10	1363
	Pooja	2020-11-05	2159
	Karan	2022-09-18	1477

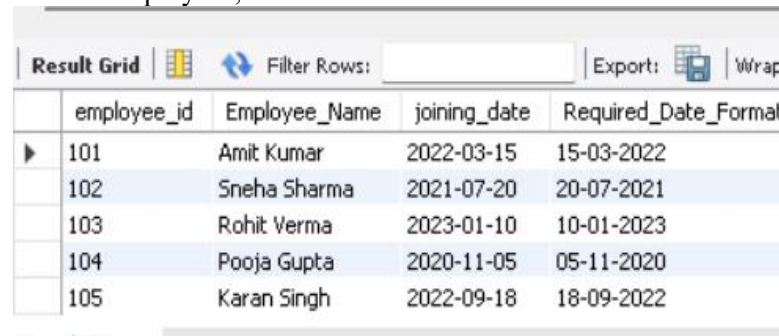
24. Add a specified number of days to a date.

```
SELECT  
    joining_date,  
    DATE_ADD(joining_date, INTERVAL 30 DAY) AS Date_After_30_Days  
FROM Employees;
```

Result Grid  Filter Rows:		
	joining_date	Date_After_30_Days
▶	2022-03-15	2022-04-14
	2021-07-20	2021-08-19
	2023-01-10	2023-02-09
	2020-11-05	2020-12-05
	2022-09-18	2022-10-18

25. Display employees' joining dates in a required date format.

```
SELECT
    employee_id,
    CONCAT(first_name, ' ', last_name) AS Employee_Name,
    joining_date,
    DATE_FORMAT(joining_date, '%d-%m-%Y') AS Required_Date_Format
FROM Employees;
```



	employee_id	Employee_Name	joining_date	Required_Date_Format
▶	101	Amit Kumar	2022-03-15	15-03-2022
	102	Sneha Sharma	2021-07-20	20-07-2021
	103	Rohit Verma	2023-01-10	10-01-2023
	104	Pooja Gupta	2020-11-05	05-11-2020
	105	Karan Singh	2022-09-18	18-09-2022

Note: Function names can vary slightly between MySQL, Oracle and SQL Server. Since you use MySQL Workbench, these can be framed specifically for MySQL.

Submission:

1. Take screenshots of each step's output.
2. Save the SQL scripts in a text file.